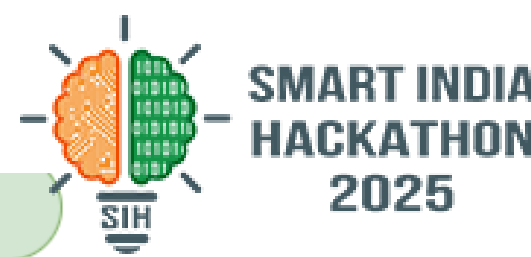


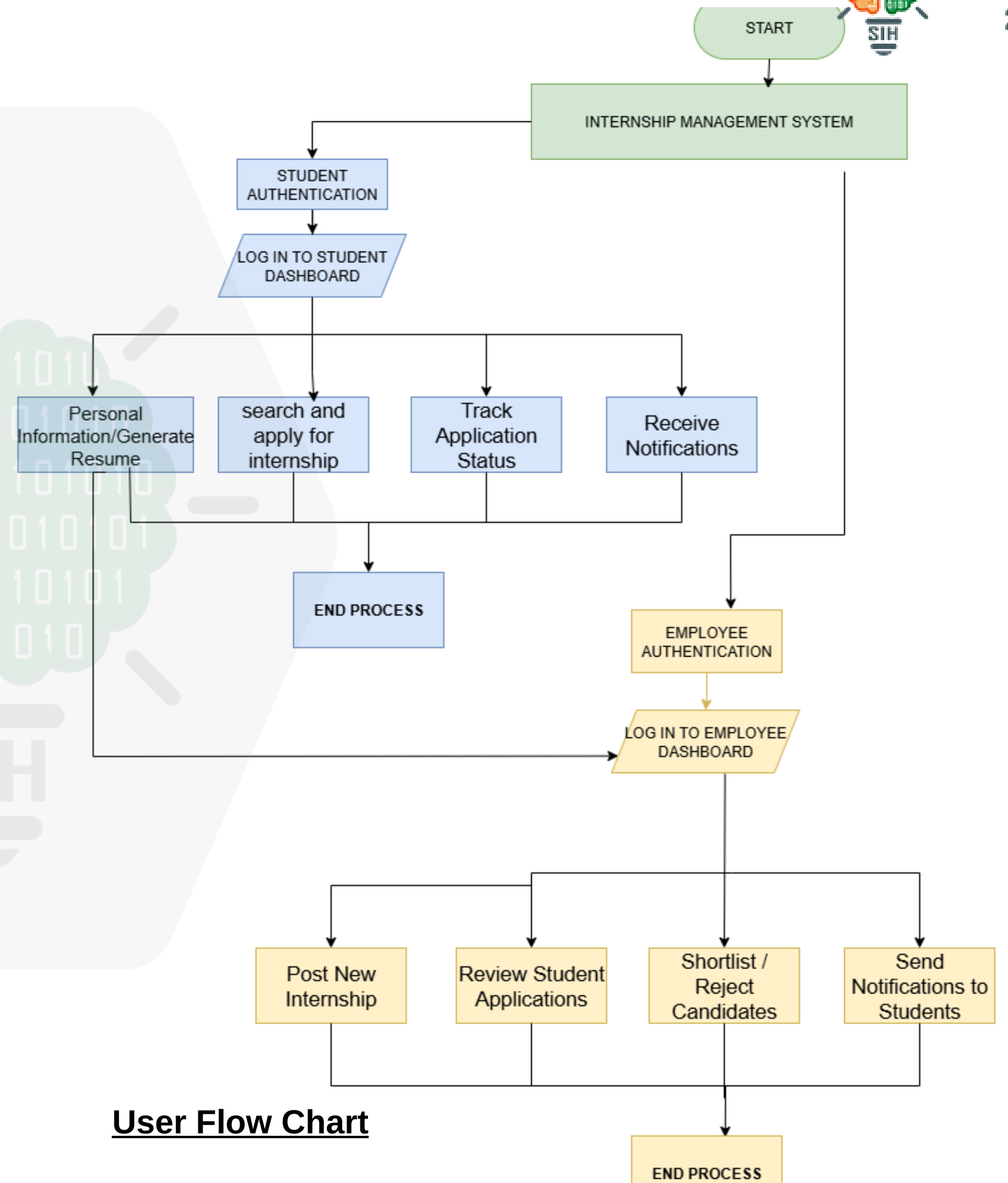
SMART INDIA HACKATHON 2025



YuvaSetu

- Problem Statement ID - [SIH25034](#)
- Problem Statement Title - [AI-Based Internship Recommendation Engine for PM Internship Scheme](#)
- Theme - [Smart Education](#)
- PS Category - [Software](#)
- Team ID - [101092](#)
- Team Name - [Team Havoc](#)
- Team members -

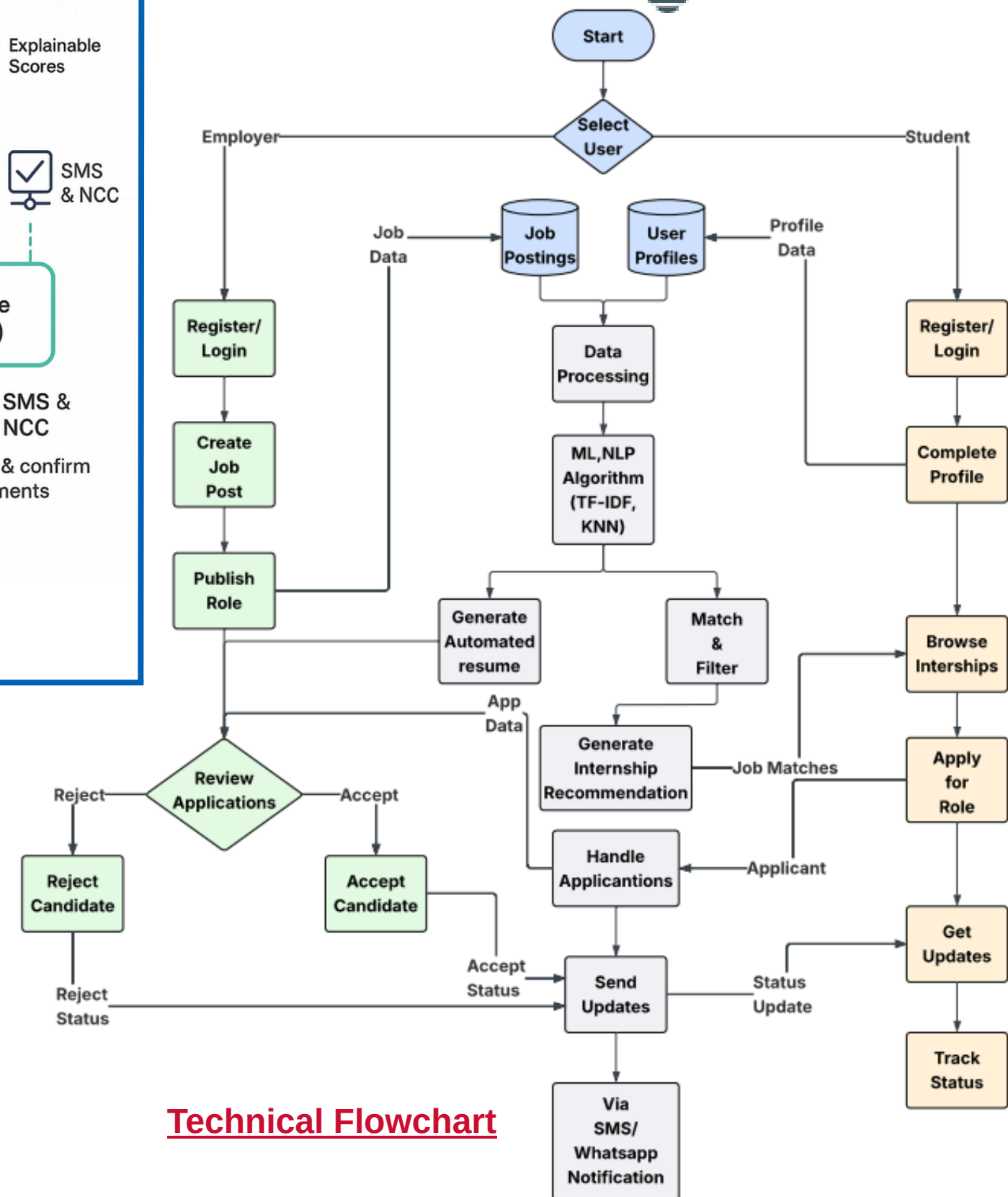
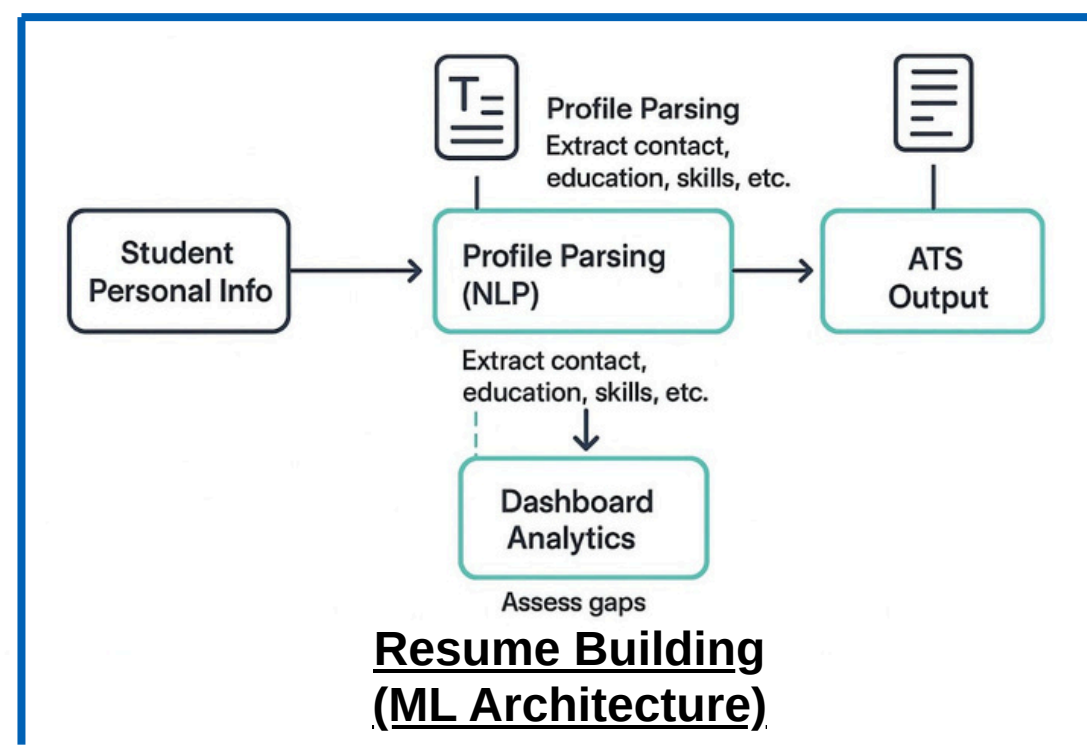
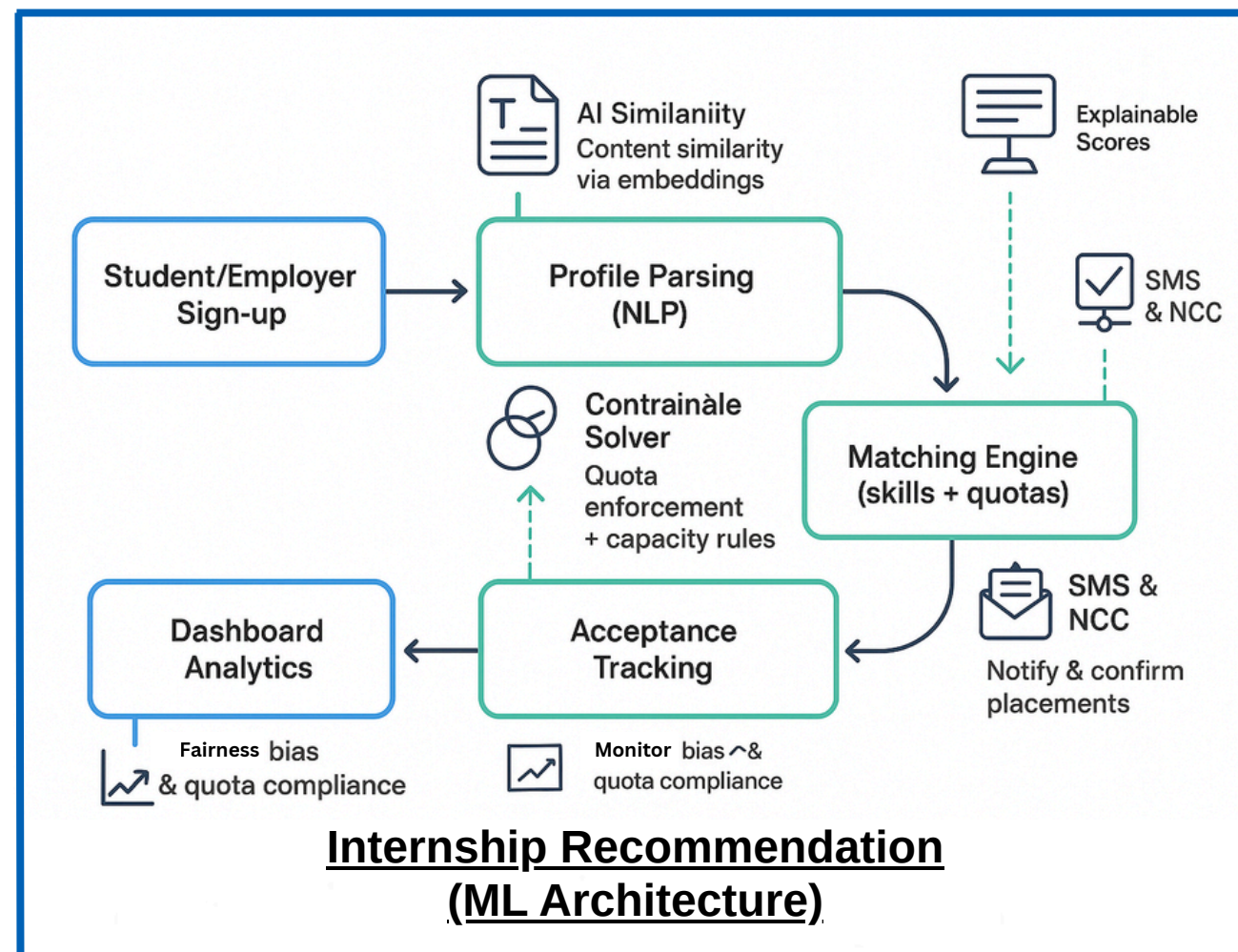
Joy Banerjee	Sharon Moses
Nashrah Ansari	Atharva Chavan
Shubham Khilari	Anamika lakhera



User Flow Chart

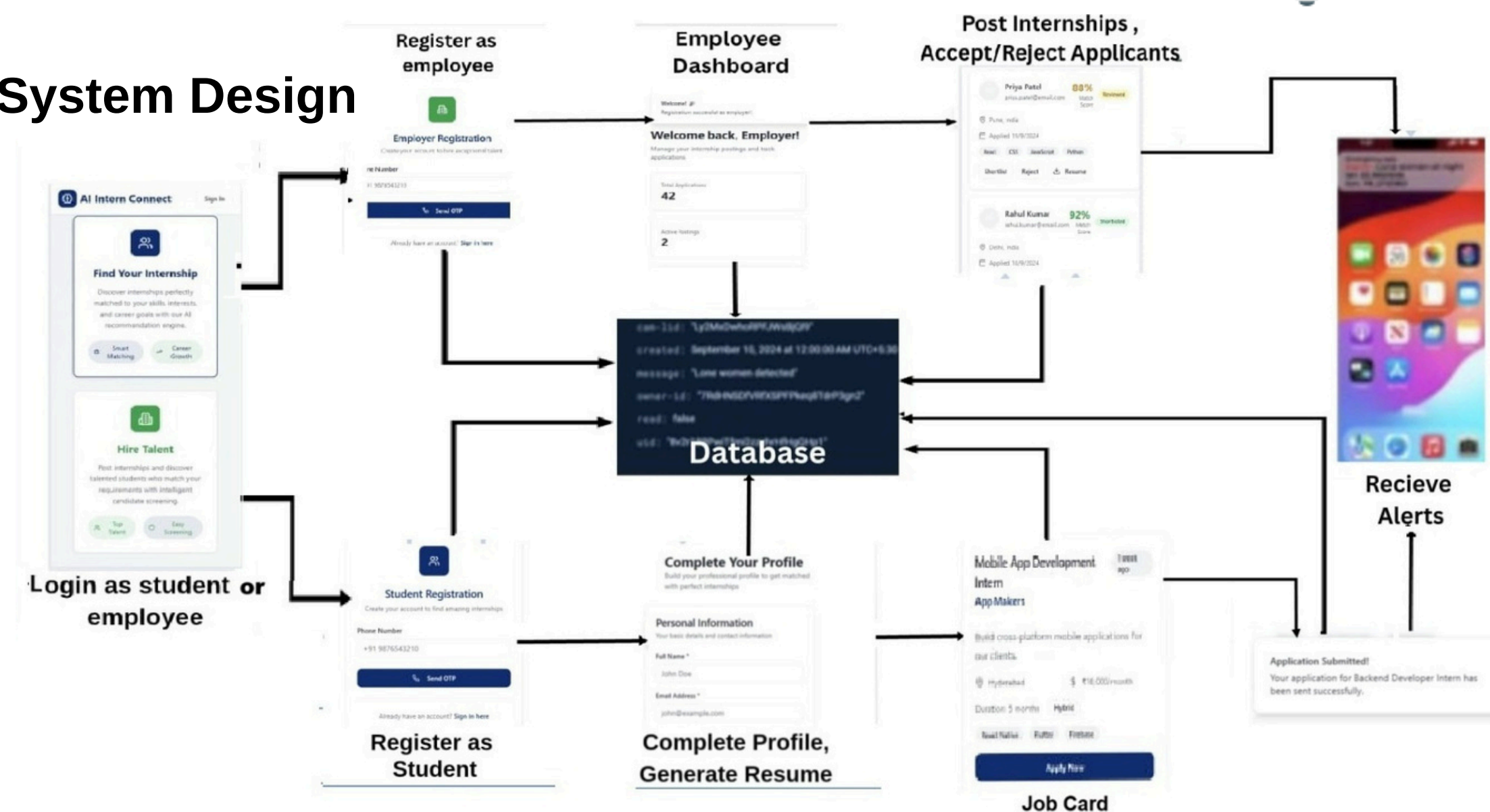
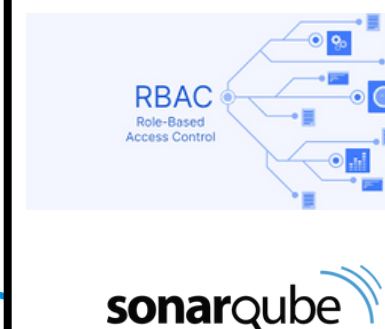
Innovation and Uniqueness

- **Explainable AI:** Shows "Why matched" with score breakdown to maximize user trust.
- **RBAC Core Security:** Governs all data access and visibility as a fundamental system constraint.
- **Automated Resume Generation:** Creates tailored resumes from profile data, ending the document upload hurdle.
- **Cascading Offers:** Automatically reallocates declined slots to next-best candidates, boosting placement efficiency.
- **Fairness-First Monitoring:** Uses demographic data (KPIs) to actively detect and reduce bias in matching decisions.



System design Explanation:

- **Zero-Friction Profile:** Automatically generates a tailored resume from structured user input, eliminating the need for document uploads.
- **ML-Light Recommendation Core:** Uses TF-IDF and k-NN to calculate relevance, ensuring the engine is fast, resource-efficient, and simple.
- **Focused Output Control:** Displays only 3 to 5 highly relevant job suggestions on the dashboard, directly solving candidate information overload.
- **Transparent Tracking & Trust:** Enables full application status tracking and features an Explainable AI Score to build candidate trust.
- **Secure, Multi-Modal Engagement:** RBAC (Role-Based Access Control) governs all access, and critical updates are sent via Real-time Notifications (SMS and WhatsApp) to guarantee outreach to rural users.

System Design**Tech Stack:****Frontend****Backend****Deployment****Code Quality & Security**

FEASIBILITY AND VIABILITY

FEASIBILITY

Feasibility Analysis

Technical Feasibility:

1. Built using open-source tools like FastAPI, React, PostgreSQL efficiently.
2. Backend scalable with PostgreSQL managing profiles, vacancies, and notifications.

Economic Feasibility:

1. Open-source stack lowers software costs and cloud hosting expenses.
2. Cloud pay-as-you-go lowers deployment and scaling financial burdens.

CHALLENGES AND SOLUTIONS

Manual and Opaque Matching
Traditional matching is slow, biased, and lacks transparency, leading to unfair placements and high dropout rates.

Introduce an AI-driven matching engine (hybrid ML/NLP + constraint solvers) that provides transparent "Match Scores" with explainability.

High candidate and vacancy volumes make manual allocation slow and inefficient.

Automate the process using scalable, cloud-based systems with batch and async processing

Skills, quotas, and location constraints are often ignored or poorly applied.

Build a policy-aware engine that enforces quota logic, skill matching, and fairness rules.

Poor data quality and lack of verification reduce reliability and trust in the system.

Standardize inputs with structured forms and enable lightweight verification via colleges or digital proofs.

IMPACT AND BENEFITS

Potential Impact on the Target Audience

- Inclusive Opportunities – Ensures fair allocation of internships through AI-driven matching with quotas for aspirational districts, women, and marginalized groups.
- Improved Placement Success – Matches candidate skills and employer requirements more accurately, reducing mismatches and dropout rates.
- Data-Driven Decisions – Real-time dashboards provide insights on placements, vacancies, and fairness metrics for administrators and policymakers.
- Enhanced Engagement – Continuous notifications (SMS/Email/WhatsApp) keep students and employers actively involved in the process.

Benefits of the Solution

- Scalable & Sustainable – Cloud-based design allows expansion to millions of users while keeping costs low.
- Efficient & Transparent – Automates matching, allocation, and reporting with explainable match scores and audit logs.
- Trust & Fairness – Builds credibility by enforcing reservation policies and providing transparent selection.
- Career Empowerment – Widens access to quality internships, especially for students from rural or underrepresented backgrounds, boosting employability and long-term growth.

RESEARCH AND REFERENCES

1

Cosine Similarity TF-IDF accuracy.
text classification



Purpose: This validates your core technical decision to use Cosine Similarity over standard metrics (like Euclidean Distance) to ensure the matching is based purely on the conceptual relevance of skills, which is the most accurate approach for text vectors.

2

k-Nearest Neighbors fast retrieval
information filtering



Purpose: Justifies k-NN as an "ML-Light" choice that is highly efficient for fast retrieval and ranking, aligning with the project's requirement for a non-resource-intensive system.

3

Constrained Optimization Integer
Programming fair allocation policy.



Purpose: This is the formal academic basis for your "Policy-Aware Matching" and "Fair Allocation" claims. It mathematically justifies embedding policy requirements (like quotas or aspirational district targets) as core constraints in the final solution set, proving a sophisticated approach to fairness.

4

Explainable AI (XAI) multi-criteria
decision making transparency.



Purpose: Validates your innovative feature of providing an Explainable Score Breakdown (e.g., Skills 70%, Location 30%). It shows you've considered the human factor of trust and transparency, a key concern for the target audience of first-generation learners.

5

Multi-Modal Notification Systems
rural access low bandwidth



Purpose: Supports your practical strategy of using SMS and WhatsApp for updates. Research in this area confirms that relying on standard internet (email/in-app) is unreliable for rural populations, validating your choice of robust, low-bandwidth communication channels.